K. J. Somaiya School of Engineering, Mumbai-77

Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 8

TITLE : Refinement of the interface of L Y project application.

Exp_06AIM : To implement refinement process for the interface of LY project

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Expected OUTCOME of Experiment:

CO 4: Refinement, Prototyping, Implementation of product/ service.

Books/ Journals/ Websites referred:

1 “Design Thinking”, Gavin Ambrose Design Paul Harris

Theory:

- Designing is an iterative process
- With every iteration; appearance, performance of the system may change
- Working up a design idea involves the continued refinement of the artwork and the message it communicates.
- Refinement sees small yet significant changes made to a design in order to enhance the idea and increase the effectiveness of its ability to communicate.

- Refinement can be done in :
  - Thinking in images and signs
  - Appropriation
  - Modification
  - Thinking in shapes and colours

Implementation:

Details about all steps of the Design Process.
1. Define

- **Problem:** Students face difficulty finding and tracking scholarships; Trusts need verified applicants; Admin must ensure transparency.
- **Target users:** Students, Trusts/NGOs, SuperAdmin.
- **Goals:** Simple registration, secure KYC, transparent application tracking, easy approval/rejection, analytics dashboards.

2. Research

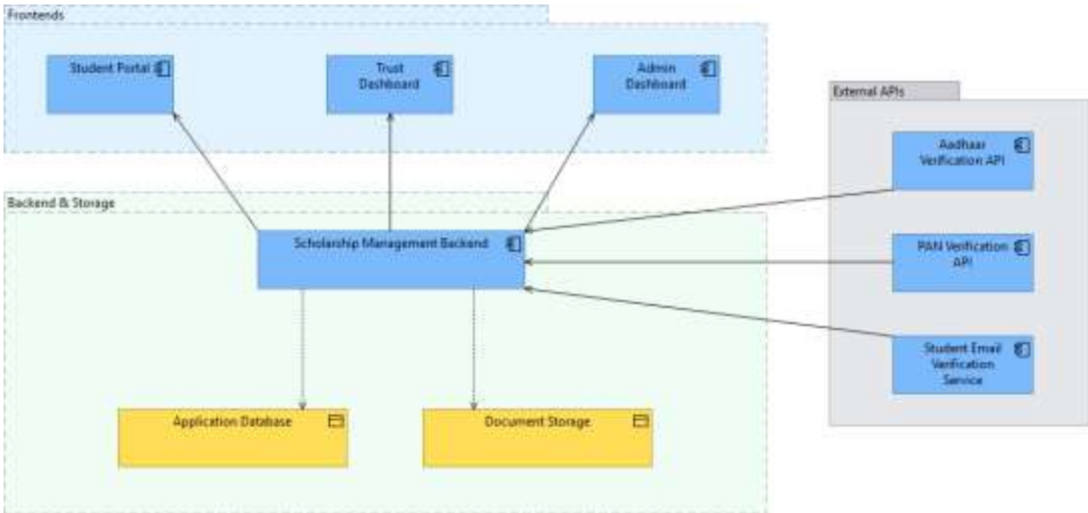
- Studied similar platforms (e.g., NSP Portal, private scholarship sites).
- Identified pain points: lengthy forms, lack of real-time status, unclear trust verification.
- Collected feedback from potential student users → need for **mobile-friendly, simple dashboards**.

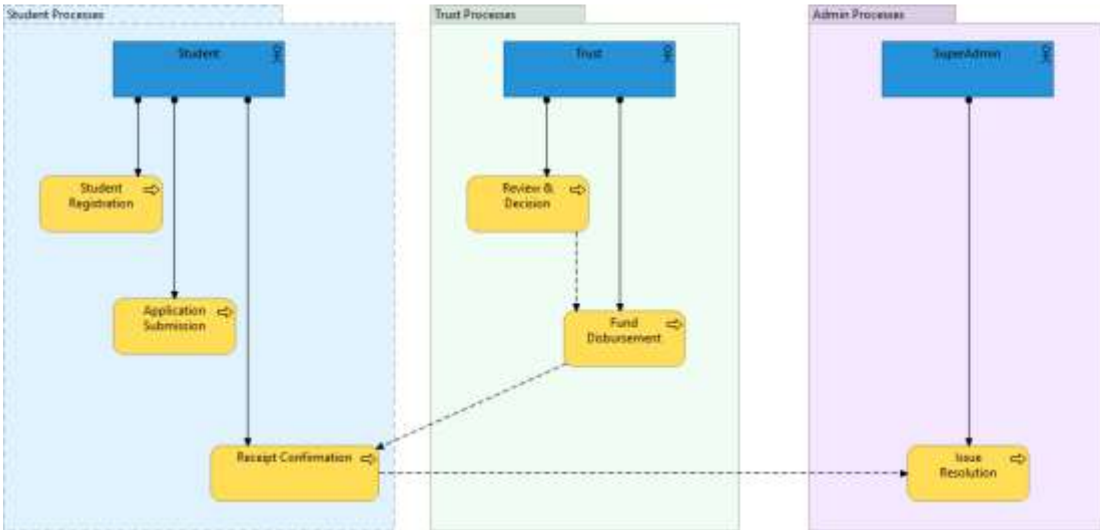
3. Ideate

- Brainstormed layouts:
  - Student Dashboard: Profile, Apply Form, Application Status, Receipts.
  - Trust Dashboard: Pending Applications, History, Analytics.
  - Admin Dashboard: Trust verification, Student/Trust blacklist, Dispute resolution, Reports.

4. Prototype

- Low-fidelity wireframes in Figma (login/signup, student application form, dashboards).
- Navigation flow: **Login** → **Role-based Dashboard** → **Actions (Apply/Approve/Manage)**.
- Integrated OTP verification and document upload interface.





5. Select

- Choose **Client-Server** Architecture.
- Choose **responsive card-based UI** with TailwindCSS for clarity.
- Choose **role-based dashboards** to keep workflows separate and secure.
- Simple **stepper-based application form** for students (personal → education → family → bank → upload).

6. Implement

- Tools: React.js (frontend), Express.js (backend), TailwindCSS (UI), PostgreSQL (DB).
- Authentication: JWT + OTP for email verification.
- Secure upload (Cloudinary/S3) for documents.
- Dark/light mode for accessibility.

7. Learn

- Continuous feedback from test users (students, NGO volunteers).
- Metrics: completion rate of applications, average approval time, dispute resolution rate.
- Iterative improvements: simplifying forms, better mobile optimization, adding chatbot support.

Abstract about LY project application:

ScholarBridge is a web platform that connects **students in need of scholarships** with **trusts/NGOs providing funds**, managed by a **SuperAdmin**.

- **Students**: Register with verified student email, upload KYC docs (Aadhaar), fill in educational, family, and financial details, apply once per year, and track funding status.
- **Trusts/NGOs**: After admin approval, can view applications, approve/reject requests, and manage fund disbursement history.

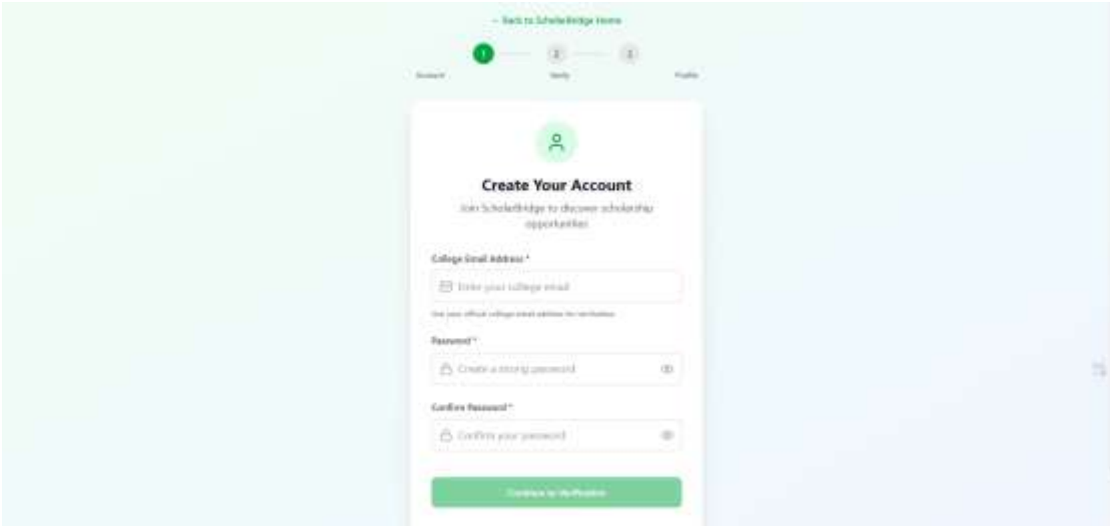
SuperAdmin: Manages students and trusts, resolves disputes, blacklists fraud accounts, and provides analytics.

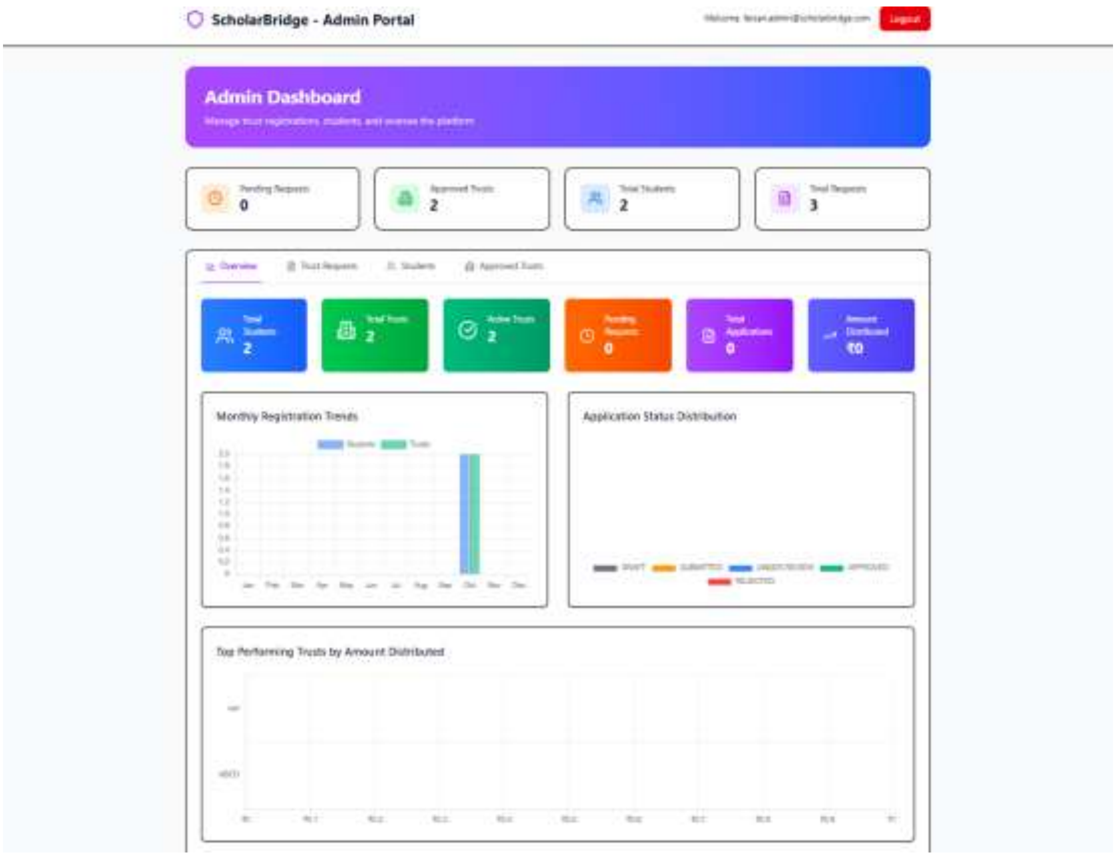
Existing Interface:

Refinement with respect to:

- Thinking in images and signs
- Appropriation
- Modification
- Thinking in shapes and colours

Refined Interface:


Create Your Account


ScholarBridge - Admin Portal
Admin Dashboard
Monthly Registration Trends
Application Status Distribution
Top Performing Trusts by Amount Distributed

Made the UI and UX better looking.
Added analytics in the admin dashboard for better analysis of the website usage.

<u>Post Lab Descriptive Questions</u>

1. Explain the concept of appropriation in Refinement.

Appropriation in the refinement phase involves reusing existing concepts, designs, or elements and reshaping them to suit a different purpose, context, or interpretation. It means drawing inspiration from prior works and transforming them to develop something more polished or relevant. During this stage, creators refine their ideas by blending recognizable components with inventive changes, allowing them to build on established solutions, streamline the creative process, and produce results that are both imaginative and impactful.

Date: 19/10/25 **Signature of faculty in-charge**